

SingleRAN

LTE FDD and NR SUL Dynamic Spectrum Sharing Feature Parameter Description

Issue

Draft B

Date

2026-01-30



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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 SRAN22.1 Draft B (2026-01-30)

This issue includes the following changes.

Technical Changes

None

Editorial Changes

Corrected the descriptions of the mutually exclusive relationship of LTE FDD and NR Uplink Spectrum Sharing with DSS preferential-resource percentage adaptation. For details, see [3.3.2.2 Mutually Exclusive Functions](#).

Revised descriptions in this document.

1.2 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 03 (2025-11-20).

Technical Changes

Change Description	Parameter Change	Base Station Model
Added support for LTE FDD and NR Uplink Spectrum Sharing by the UMPThb. For details, see 3.3.3 Hardware .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R - DBS3900 LampSite and DBS5900 LampSite
Added incompatibility of BTS5929R with LTE FDD and NR Uplink Spectrum Sharing. For details, see 3.3.3 Hardware .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Revised the constraints affecting cell activation due to license insufficiency. For details, see 3.3.1 Licenses .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added a mutually exclusive relationship of LTE FDD and NR Uplink Spectrum Sharing with LTE flexible spectrum. For details, see 3.3.2.2 Mutually Exclusive Functions .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
Added a mutually exclusive relationship of LTE FDD and NR Uplink Spectrum Sharing with base station equipment reliability. For details, see 3.3.2.2 Mutually Exclusive Functions .	None	5900 series base stations (macro base stations) excluding the BTS5929R

Editorial Changes

Revised descriptions in this document.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to LTE FDD and NR.

2.3 Features in This Document

This document describes the following features.

RAT	Feature ID	Feature Name	Chapter/Section
LTE FDD	MRFD-13122 3	LTE FDD and NR Uplink Spectrum Sharing (LTE FDD)	3 LTE FDD and NR Uplink Spectrum Sharing
NR TDD	MRFD-13126 3	LTE FDD and NR Uplink Spectrum Sharing (NR)	

2.4 Differences

Table 2-1 Differences between NR FDD and NR TDD

Function Name	Difference	Chapter/Section
LTE FDD and NR Uplink Spectrum Sharing	Supported only in NR TDD	3 LTE FDD and NR Uplink Spectrum Sharing

Table 2-2 Differences between NSA and SA

Function Name	Difference	Chapter/Section
LTE FDD and NR Uplink Spectrum Sharing	None	3 LTE FDD and NR Uplink Spectrum Sharing

Table 2-3 Differences between high frequency bands and low frequency bands

Function Name	Difference	Chapter/Section
LTE FDD and NR Uplink Spectrum Sharing	Supported only in low frequency bands	3 LTE FDD and NR Uplink Spectrum Sharing

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

3

LTE FDD and NR Uplink Spectrum Sharing

3.1 Principles

5G is usually deployed in C-band in the early stage of network construction. The 5G networks deployed in C-band face the following two issues:

- C-band's uplink coverage is inferior to its downlink coverage, and that affects the deployment of 5G networks.
- Uplink user experience is unsatisfactory, as 5G networks deployed in C-band adopt the time division multiplexing (TDM) mode, and the quantity of time-frequency resources used for uplink data transmission is limited.

To address these issues, the following NR functions have been developed:

- **UL and DL Decoupling:** This function can be used in areas with limited NR uplink coverage to improve NR uplink coverage by allowing downlink data to be transmitted in C-band and uplink data to be transmitted in a sub-3 GHz frequency band. For details about UL and DL Decoupling, see *UL and DL Decoupling*.
- **Super Uplink:** This function improves NR uplink experience by allowing uplink data transmission in a sub-3 GHz frequency band and in C-band in different timeslots, thereby achieving uplink data transmission in all timeslots. For details about this function, see *Super Uplink*.

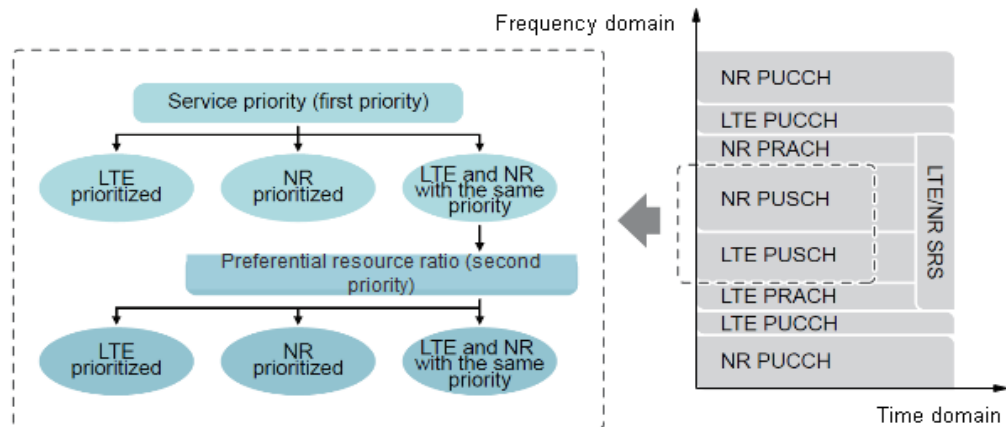
If operators do not have dedicated sub-3 GHz uplink spectrum for 5G, neither of these functions can be used. LTE FDD and NR Uplink Spectrum Sharing is introduced to allow for the sharing of LTE FDD uplink low-band spectrum resources between LTE FDD cells and NR supplementary uplink (SUL) cells. The uplink low-band spectrum sharing serves as a basis for UL and DL Decoupling and Super Uplink on the NR TDD side.

For ease of description in this document, LTE is used to refer to LTE FDD, for example, LTE cells refer to LTE FDD cells. The NR cells in this function refer to NR SUL cells.

Technical Principles

With LTE and NR uplink spectrum sharing, LTE and NR uplink physical channels share spectrum resources based on the service priority (negotiated between the eNodeB and gNodeB) and spectrum sharing preferential resource ratio (specified by the **LteNrSpctShrCellGrp.LteNrSpctShrLtePriResRatio** parameter), as shown in **Figure 3-1**. For details about LTE and NR service priorities, see *Uplink Scheduling in eRAN Feature Documentation* and *QoS Management in 5G RAN Feature Documentation*.

Figure 3-1 LTE and NR uplink spectrum sharing



This function implements:

- Dynamic spectrum allocation to sounding reference signals (SRSs) between LTE and NR in the time domain based on the traffic volumes of LTE and NR networks to improve the resource usage.
- Coordinated scheduling of uplink physical channel resources to avoid interference during LTE and NR uplink spectrum sharing.

Coordinated scheduling of uplink physical channel resources can be performed only when LTE and NR subframes are aligned. Alignment is configured by the LTE parameter **CellFrameOffset.FrameOffset** or **ENodeBFrameOffset.FddFrameOffset** and the NR parameter **gNodeBParam.FrameOffset** or **gNBFreqBandConfig.FrameOffset**. If both LTE parameters are configured, the value of the **CellFrameOffset.FrameOffset** parameter is used. If both NR parameters are configured, the value of the **gNBFreqBandConfig.FrameOffset** parameter is used. When LTE and NR subframes are aligned, the uplink physical channel resources are scheduled as follows:

- NR PUCCH: RBs do not need to be reserved for the NR PUCCH if channel decoupling is enabled on the NR side. Ten RBs that are symmetrically configured at the two ends of the bandwidth are reserved for the NR PUCCH if channel decoupling is not enabled on the NR side. For details about channel decoupling, see *Super Uplink* and *UL and DL Decoupling*.
- LTE PUCCH: The LTE PUCCH is symmetrically configured adjacent to the NR PUCCH at the two ends of the bandwidth. The start position of the LTE

PUCCH at the lower end and the end position of the LTE PUCCH at the upper end can be configured through the **CellRbReserve.RbRsvMode**, **CellRbReserve.RbRsvType**, **CellRbReserve.RbRsvStartIndex**, and **CellRbReserve.RbRsvEndIndex** parameters. The LTE PUCCH can also be deployed by the eNodeB at the positions described in [Table 3-1](#). The number of RBs occupied by the LTE PUCCH is parameter-specified or is dynamically adjusted. For details, see the descriptions of the PUCCH in *Physical Channel Resource Management* in *eRAN Feature Documentation*.

Table 3-1 Start and end positions of the LTE PUCCH

Bandwidth	Start RB Index at the Lower End	End RB Index at the Upper End
10 MHz	4	45
15 MHz	3	71
20 MHz	2	97

Assume that 12 RBs are configured for the LTE PUCCH in 15 MHz bandwidth scenarios. Then, the LTE PUCCH occupies RBs with indexes 3 to 8 at the lower end, and occupies RBs with indexes 66 to 71 at the upper end.

NOTICE

If the LTE or NR cell is not deactivated before the preceding configurations are performed:

- When the **FORBID_PARA_CFG_VERIFY_SW** option of the **eNodeBResModeAlgo.ParamCfgProtectSw** parameter is deselected, a command to reconfigure **CellRbReserve.RbRsvMode**, **CellRbReserve.RbRsvStartIndex**, or **CellRbReserve.RbRsvEndIndex** will fail to be executed.
- When the **FORBID_PARA_CFG_VERIFY_SW** option of the **eNodeBResModeAlgo.ParamCfgProtectSw** parameter is selected, the NR cell may become out of service.

Evaluate the preceding risks and determine whether to deactivate the LTE and NR cells before performing the preceding configurations.

- **LTE PRACH:** The LTE PRACH always occupies 6 RBs. It is recommended that RBs adjacent to the LTE PUCCH at the lower end be reserved for the LTE PRACH. The frequency-domain position of RBs can also be parameter configured. For details, see descriptions of the PRACH frequency-domain position in *Random Access Control* in *eRAN Feature Documentation*. The LTE PRACH frequency-domain position must not overlap with the frequency-domain positions of the LTE PUCCH, NR PUCCH, or NR PRACH.
- **NR PRACH:** RBs do not need to be reserved for the NR PRACH if channel decoupling is enabled on the NR side. Six RBs are reserved for the NR PRACH if channel decoupling is not enabled on the NR side. The NR PRACH is configured adjacent to the LTE PUCCH at the upper end. The frequency-

domain start position of the NR PRACH is determined based on the LTE PUCCH.

- LTE and NR PUSCHs: The total resources available for LTE and NR PUSCHs equal the total available spectrum resources minus the resources occupied by LTE and NR PUCCHs and PRACHs. The total available PUSCH resources for LTE and NR are dynamically shared between LTE and NR based on traffic volume requirements.

The preceding allocation principles determine the total spectrum resources for LTE and NR SUL cells. The spectrum resources occupied by the uplink of the LTE and NR SUL cells can be obtained by observing the counters listed in [Table 3-2](#).

Table 3-2 Counters for measuring the numbers of uplink available RBs in LTE and NR SUL cells

Counter ID	Counter Name
1526759050	L.ChMeas.PR.B.UL.Actual.Avail
1526759052	L.ChMeas.PR.B.UL.PUSCH.Actual.Avail
1911827161	N.PR.B.UL.Actual.Avail.Avg
1911827162	N.PR.B.UL.PUSCH.Actual.Avail.Avg

Function Configuration

This function requires that LTE and NR be configured with the same clock source and be time synchronized. The function switches and related parameters on the two sides must be configured.

The clock source and time synchronization mode are configured through LTE parameters **TASM.MODE**, **TASM.CLKSRC**, **PEERCLK.PN**, **PEERCLK.PNT**, **PEERCLK.PS**, and **TASM.CLKSYNCMODE**. The NR SUL cell is already time synchronized with the NR C-band cell. Therefore, in separate-MPT LTE/NR scenarios, when the clock sources configured for the LTE and NR sides are different, the clock source for the LTE side must be changed to be the same as that for the NR side, and time synchronization must be achieved. In co-MPT LTE/NR scenarios, the LTE and NR sides already share the clock source and are time synchronized, and reconfiguration is not required. If the requirements are not met, the NR SUL cell fails to be activated.

On the LTE side, set the **SpectrumCloud.SpectrumCloudSwitch** parameter to **LTE_NR_UPLINK_SPECTRUM_SHR** to enable uplink spectrum sharing. In addition, perform the following configurations:

- Add LTE cells to an LTE spectrum sharing cell group by configuring the **SpectrumCloud.LocalCellId** and **SpectrumCloud.LteNrSpectrumShrCellGrpId** parameters.
- Configure super uplink spectrum sharing, which is controlled by the **LNR_SPECTRUM_SHR_SUPERUL_SW** option of the **SpectrumCloud.SpectrumCloudEnhSwitch** parameter.
- Configure the percentage of spectrum resources to be preferentially allocated to LTE services through the

LteNrSpctShrCellGrp.LteNrSpctShrLtePriResRatio parameter. The percentages of resources which are preferentially allocated to LTE and NR can be set based on their traffic volume requirements. This ensures that resources are allocated properly.

- Configure the LTE and NR co-carrier co-CPRI data switch through the **LteNrSpctShrCellGrp.LteNrCoCarrCoCpriDataSw** parameter. This parameter can be set to **ON** only when LTE and NR use the CPRI MUX topology. When this parameter is set to **ON**, the requirements on the CPRI bandwidth are reduced.

On the NR side, select the **LTE_NR_UL_SPECTRUM_SHARING_SW** option of the **NRDUCellAlgoSwitch.SpectrumCloudSwitch** parameter to enable this function. In addition, perform the following configurations:

- Configure the association between the LTE spectrum sharing cell group and the NR spectrum sharing cell group through the **gNBDDLteNrSpctShrCg.NrSpctShrCellGrpId** and **gNBDDLteNrSpctShrCg.LteSpctShrCellGrpId** parameters.
- Add NR SUL cells to the NR spectrum sharing cell group by setting the **NRDUCellSpctCloud.NrDuCellId** and **NRDUCellSpctCloud.NrSpctShrCellGrpId** parameters.

In addition to the preceding parameter configurations, you also need to set the parameters in [Technical Principles](#).

3.2 Network Analysis

3.2.1 Benefits

- This function together with UL and DL Decoupling improves the NR uplink throughput via the establishment of SUL cells, and helps achieve fast NR deployment. For details about the benefits provided by UL and DL Decoupling, see *UL and DL Decoupling*.
- This function together with the Super Uplink function improves NR uplink service experience via the establishment of SUL cells. For details about the benefits provided by Super Uplink, see *Super Uplink*.

3.2.2 Impacts

The impacts between this function and other functions are described in [3.3.2.3 Function Impacts](#).

This function affects only LTE networks. It has no impact on NR networks. This function reduces the number of uplink RBs available for the LTE PUSCH, decreasing the uplink throughput of LTE UEs. The greater the uplink load, the larger this impact.

Uplink throughput of LTE UEs = $(L.Thrp.bits.UL - L.Thrp.bits.UE.UL.SmallPkt) / L.Thrp.Time.UE.UL.RmvSmallPkt$. [Table 3-3](#) lists the related counters.

Table 3-3 Counters used for measuring the uplink throughput of LTE UEs

Counter ID	Counter Name
1526728259	L.Thrp.bits.UL
1526729415	L.Thrp.bits.UE.UL.SmallPkt
1526729416	L.Thrp.Time.UE.UL.RmvSmallPkt

3.3 Requirements

3.3.1 Licenses

This function requires both feature and capacity licenses. [Table 3-4](#) and [Table 3-5](#) list the required feature licenses. [Table 3-6](#) and [Table 3-7](#) list the required capacity licenses.

Table 3-4 Feature licenses (for 3900 and 5900 series base stations)

RAT	Feature ID	Feature Name	Model	Description	NE	Sales Unit
LTE FDD	MRFD-1 31223	LTE FDD and NR Uplink Spectrum Sharing (LTE FDD)	LT1S0FN USS00	LTE FDD and NR Uplink Spectrum Sharing (LTE FDD) (per Cell)	eNode B	per Cell
NR	MRFD-1 31263	LTE FDD and NR Uplink Spectrum Sharing (NR)	NR0S0F NUSS00	LTE FDD and NR Uplink Spectrum Sharing (NR) (per Cell)	gNode B	per Cell

RAT	Feature ID	Feature Name	Model	Description	NE	Sales Unit
				For the license control item LT1S0FNUS00: • If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. • If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation: – When the license sales number pre-check function is enabled, the SpectrumCloud.SpectrumCloudSwitch parameter cannot be set to LTE_NR_UPLINK_SPECTRUM_SHR due to license insufficiency, preventing cell activation failures. – When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified. For the license control item NR0S0FNUS00: • If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. • If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation: – When the license sales number pre-check function is enabled, the LTE_NR_UL_SPECTRUM_SHARING_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter cannot be selected due to license insufficiency, preventing cell activation failures. – When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified. On the LTE side, the license sales number pre-check function is controlled by the LIC_SALE_NUM_PRE_CHECK_SW option of the eNBLicenseAlmThd.LicenseCheckCtrlSw parameter. The license sales number pre-check function is controlled by the LIC_SALE_NUM_PRE_CHECK_SW option of the gNBLicenseAlmThld.LicenseChkSw parameter.		

Table 3-5 Feature licenses (for DBS3900 LampSite and DBS5900 LampSite)

RAT	Feature ID	Feature Name	Model	Description	NE	Sales Unit
LTE FDD	MRFD-1 31223	LTE FDD and NR Uplink Spectrum Sharing (LTE FDD)	LT1SFNS PCS00	LTE FDD and NR Uplink Spectrum Sharing (LampSite FDD) (per Carrier per pRRU)	eNode B	per Carrier per pRRU
NR	MRFD-1 31263	LTE FDD and NR Uplink Spectrum Sharing (NR)	NR0SFN SPCS01	LTE FDD and NR Uplink Spectrum Sharing (LampSite NR) (per Carrier per pRRU)	gNode B	per Carrier per pRRU
For the license control item NR0SFNSPCS01: • If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. • If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation, the cell cannot be activated again after the feature is enabled and the parameters that cause automatic cell reset are modified. For the license control item LT1SFNSPCS00: • If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. • If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation: – When the license sales number pre-check function is enabled, the SpectrumCloud.SpectrumCloudSwitch parameter cannot be set to LTE_NR_UPLINK_SPECTRUM_SHR due to license insufficiency, preventing cell activation failures. – When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified. The license sales number pre-check function is controlled by the LIC_SALE_NUM_PRE_CHECK_SW option of the eNBLicenseAlmThd.LicenseCheckCtrlSw parameter.						

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

Table 3-6 Capacity licenses (for 3900 and 5900 series base stations)

RAT	Model	Description	NE	Sales Unit
LTE FDD	LT1SRFSPCS00	RF Spectrum Sharing License (FDD)	eNodeB	per Band per RU
LTE FDD	LT1S5000RFSS	Spectrum Sharing License for 5000 Series RF Module (FDD)	eNodeB	per Band per RU
NR	NR0S0RFSSL00	RF Spectrum Sharing License (NR)	gNodeB	per Band per RU
NR	NR0SSSLRFM00	Spectrum Sharing License for 5000 Series RF Module (NR)	gNodeB	per Band per RU

Table 3-7 Capacity licenses (for DBS3900 LampSite and DBS5900 LampSite)

RAT	Model	Description	NE	Sales Unit
LTE FDD	LT1SFDSPCS30	Spectrum Sharing License for LampSite pRRU (LTE FDD)	eNodeB	per Band per pRRU
LTE FDD	LT1SFDSPCS50	Spectrum Sharing License for 5000 Series LampSite pRRU (LTE FDD)	eNodeB	per Band per pRRU
NR	NR0SSPECTT00	Spectrum Sharing License for LampSite pRRU (NR)	gNodeB	per Band per pRRU
NR	NR0SSPECTT50	Spectrum Sharing License for 5000 Series LampSite pRRU (NR)	gNodeB	per Band per pRRU

 NOTE

The insufficiency of certain capacity licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

Customers must purchase the Spectrum Sharing License for 5000 Series RF Module when 5000 series RF modules are used. When other RF modules are used, customers must purchase the RF Spectrum Sharing License. When 5000 series RF modules and other RF modules are both used, customers must purchase both hardware licenses.

Customers must purchase the Spectrum Sharing License for 5000 Series LampSite pRRU when 5000 series LampSite pRRUs are used. When non-5000 series LampSite pRRUs are used, customers must purchase the Spectrum Sharing License for LampSite pRRU. When 5000 series LampSite pRRUs and non-5000 series LampSite pRRUs are both used, customers must purchase both hardware licenses.

3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

3.3.2.1 Prerequisite Functions

- Prerequisite functions on the LTE side

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Normal CP	Cell.UlCyclicPrefix	<i>Extended CP</i>	Set this parameter to NORMAL_CP .
LTE FDD	SRS Configuration Indicator	SRSCfg.SrsConfigInd	<i>Physical Channel Resource Management</i>	Set this parameter to BOOLEAN_TRUE .
LTE FDD	FDD SRS Configuration Mode	SrsCfg.FddSrsCfgMode	<i>Physical Channel Resource Management</i>	Set this parameter to DEFAULTMODE .
LTE FDD	SRS subframe configuration	SrsCfg.SrsSubframeCfg	<i>Physical Channel Resource Management</i>	Set this parameter to SC3, SC7, SC14, or SC0 .

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Dynamic adjustment to cell-specific SRS subframe configuration	SrsSubframeRecfSwitch option of the CellAlgoSwitch.SrsAlgoSwitch parameter	<i>Physical Channel Resource Management</i>	Select the option of this parameter.
LTE FDD	Uplink adaptive HARQ	CellUlschAlgo.AdaptHarqSwitch	<i>Uplink Scheduling</i>	Set this parameter to ADAPTIVE_HARQ_SW_ON or ADAPTIVE_HARQ_SW_SEMI_ON .

- Prerequisite functions on the NR side
None

3.3.2.2 Mutually Exclusive Functions

- Mutually exclusive functions on the LTE side

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LTE FDD	LTE flexible spectrum	Cell.CustomizedBandwidthCfgInd set to FLEX_CFG	<i>LTE Flexible Spectrum (FDD)</i>
LTE FDD	RuralLink antenna solution (LTE)	None	<i>RuralLink Solution</i>
LTE FDD	SFN cell	Cell.MultiRruCellMode	<i>SFN</i>
LTE FDD	High speed mobility	Cell.HighSpeedFlag	<i>High Speed Mobility</i>
LTE FDD	Ultra high speed mobility	Cell.HighSpeedFlag	<i>High Speed Mobility</i>
LTE FDD	Compact bandwidth	Cell.CustomizedBandwidthCfgInd	<i>Compact Bandwidth (FDD)</i>

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LTE FDD	GSM and LTE Spectrum Concurrency	SpectrumCloud.SpectrumCloudSwitch set to GL_SPECTRUM_CONCURRENCY	<i>GSM and LTE Spectrum Concurrency</i>
LTE FDD	Flexible bandwidth based on overlapping carriers	DdCellGroup.DdBandwidth	<i>Flexible Bandwidth based on Overlap Carriers (FDD)</i>
LTE FDD	GSM and LTE Buffer Zone Optimization	AvoidGeranInterfSwitch option of the CellAlgoSwitch.AvoidInterfSwitch parameter	<i>GSM and LTE Buffer Zone Optimization</i>
LTE FDD	UL Refarming Zero Bufferzone	AvoidUtranInterfSwitch option of the CellAlgoSwitch.AvoidInterfSwitch parameter	<i>UL Refarming Zero Bufferzone</i>
LTE FDD	UMTS and LTE Zero Bufferzone	UMTS_LTE_ZERO_BUFFER_ZONE_SW option of the ULZeroBufferZone.ZeroBufferZoneSwitch parameter	<i>UMTS and LTE Zero Bufferzone</i>
LTE FDD	UMTS and LTE Spectrum Sharing	SpectrumCloud.SpectrumCloudSwitch set to UL_SPECTRUM_SHARING	<i>UMTS and LTE Spectrum Sharing</i>
LTE FDD	UMTS and LTE Spectrum Sharing Based on DC-HSDPA	SpectrumCloud.SpectrumCloudSwitch set to DC_HSDPA_BASED_UL_SPECTRUM_SHR	<i>UMTS and LTE Spectrum Sharing Based on DC-HSDPA</i>
LTE FDD	CDMA and LTE Zero Bufferzone	AvoidCDMAInterfSwitch option of the CellAlgoSwitch.AvoidInterfSwitch parameter	<i>CDMA and LTE Zero Bufferzone</i>
LTE FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	SpectrumCloud.SpectrumCloudSwitch set to LTE_NR_SPECTRUM_SHR	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LTE FDD	DSS preferential-resource percentage adaptation	DSS_PREFER_RES_RATIO_A DAPT_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>
LTE FDD	LTE flexible bandwidth	CellLteFlexBw.LteFlexBwSwitch	None
LTE FDD	UL CRA	UL_COORD_RES_ALLOC_SWITCH option of the ULCsAlgoPara.ULCsSw parameter	<i>Uplink Coordinated Scheduling</i>
LTE FDD	UL CPC	UL_COORD_PC_SWITCH option of the ULCsAlgoPara.ULCsSw parameter	<i>Uplink Coordinated Scheduling</i>
LTE FDD	Coordinated scheduling-based power control	CellCspcPara.CellCspcSwitch	<i>CSPC</i>
LTE FDD	PUSCH FH	CellUlschAlgo.UlHoppingType	None
LTE FDD	Flexible PUCCH configuration	PucchFlexCfgSwitch option of the CellAlgoSwitch.PucchAlgoSwitch parameter	<i>Physical Channel Resource Management</i>
LTE FDD	SRS automatic neighboring cell measurement	NCellSrsMeasPara.SrsAutoNCellMeasSwitch	None
LTE FDD	Out-of-band relay	OutOfBandRelaySwitch option of the CellAlgoSwitch.RelaySwitch parameter	<i>Relay</i>
LTE FDD	In-band relay	InBandRelayDeNbSwitch, InBandRelayReNbSwitch, and InBandRelayDistant-DeploySw option of the CellAlgoSwitch.RelaySwitch parameter	<i>Relay</i>

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)
LTE FDD	Manual blocking of PUSCH RBs	CellRbReserve.RbRsvMode set to RB_MASKING or RB_MASKING_WITH_SRS	<i>Uplink Scheduling</i>
LTE FDD	eMTC introduction	EMTC_SWITCH option of the CellEmtcAlgo.EmtcAlgoSwitch parameter	<i>eMTC</i>
NB-IoT	LTE in-band deployment	CellRbReserve.RbRsvMode set to NB_RESERVED or NB_DEPLOYMENT	<i>NB-IoT Basics (FDD)</i>
NB-IoT	NR guard band deployment	Prb.DeployMode set to GUARD_BAND and Prb.NrCellGbCfgInd set to CFG	<i>NB-IoT Basics (FDD)</i>
LTE FDD	Superior uplink coverage	CellAlgoExtSwitch.UlCoverageEnhancementSw	<i>Superior Uplink Coverage (FDD)</i>
LTE FDD	Uplink PUSCH RB reservation for MCPTT voice services	CellRbReserve.RbRsvMode set to MCPTT_RB_RESERVED	<i>PTT</i>
LTE FDD	Short TTI	SHORT_TTI_SW option of the CellShortTtiAlgo.SttiAlgoSwitch parameter	<i>Short TTI (FDD)</i>
LTE FDD	Static Multiple Beam	None	<i>Smart 8T8R (FDD)</i>
LTE FDD	Zero Guard Band Between Contiguous Intra-Band Carriers	CONTIG_INTRA_BAND_CARRIER_SW option of the ContigIntraBand-Carr.ContigIntraBand-CarrSw parameter	<i>Seamless Intra-Band Carrier Joining (FDD)</i>
LTE FDD	PIM interference cancellation	PIM_INTERF_CANCELLATION_SW option of the PimInterfSmartAvoid.PimAlgoSwitch parameter	<i>Interference Detection and Suppression</i>

- Mutually exclusive functions on the NR side

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)
NR FDD	LumiLink solution (NR)	EASY_OMNI_SITE_MODE_SW option of the NRDUfddCellTrpBeam.ScenarioModeSw parameter	<i>LumiLink Solution</i>
Low-frequency NR TDD	Service admission	QOS_FLOW_ADMISSION_SW option of the NRDUCellAlgoSwitch.RacAlgoSwitch parameter	<i>Admission Control</i>
Low-frequency NR TDD	Service preemption	QOS_FLOW_PREEMPTION_SW option of the NRDUCellAlgoSwitch.RacAlgoSwitch parameter	<i>Admission Control</i>
Low-frequency NR TDD	RB adaptation in long PUCCH formats	PUCCH_RBRES_ADAPTIVE_SWITCH option of the NRDUCellPucch.PucchAlgoSwitch parameter	<i>Channel Management</i>
Low-frequency NR TDD	SRS transmission using common PUCCH resources	COMMON_PUCCH_AVOID_SRS_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter	<i>Channel Management</i>
Low-frequency NR TDD	LTE TDD and NR spectrum sharing	LTE_NR_TDD_SPCT_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter	None
Low-frequency NR TDD	VMIMO in DAS-based indoor coverage	NRDUCellAlgoSwitch.VmimoSwitch	<i>Indoor Coverage</i>
Low-frequency NR TDD	Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios	DUAL_PLANE_TOB_SW option of the gNBRedundancy.gNBReliabilitySwitch parameter	<i>Base Station Equipment Reliability</i>

3.3.2.3 Function Impacts

- Function impacts on the LTE side

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Uplink CA	CaUl2CCSwitch option of the CaMgtCfg.CellCaAlgoSwitch parameter	<i>Carrier Aggregation</i>	The number of uplink RBs available for LTE decreases, which reduces the throughput of UEs for which FDD uplink CA takes effect.
LTE FDD	Uplink FDD +TDD CA	InterFddTddCaSwitch option of the CaMgtCfg.CellCaAlgoSwitch parameter selected and CaMgtCfg.FddTddCaUlMaxCcnNum set to 2CC	<i>Carrier Aggregation</i>	The number of uplink RBs available for LTE decreases, which reduces the throughput of UEs for which uplink FDD+TDD CA takes effect.
LTE FDD	Flexible CA	MultiCarrierFlexCaSwitch option of the CaMgtCfg.CellCaAlgoSwitch parameter	<i>Carrier Aggregation</i>	LTE FDD and NR Uplink Spectrum Sharing may affect the selected CA groups.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Downlink massive CA	• CaDL5CCSwitch option of the CaMgtCfg.CellCaAlgoSwitch parameter • UeBandCombReducedR13Sw option of the ENodeBAlgoSwitch.CaAlgoExtSwitch parameter • DLMassiveCaSwitch option of the CaMgtCfg.CellCaAlgoSwitch parameter	<i>Carrier Aggregation</i>	The LTE cells enabled with LTE FDD and NR Uplink Spectrum Sharing are not recommended as PCells. If these cells act as PCells, the PUCCH overhead is so large that SRSs cannot be configured. Consequently, the LTE network throughput decreases.
LTE FDD	LTE spectrum coordination	SpectrumCoordinationSwitch option of the CaMgtCfg.CellCaAlgoSwitch parameter	<i>LTE Spectrum Coordination</i>	The number of uplink RBs available for LTE decreases, which reduces the proportion of UEs for which LTE spectrum coordination takes effect.
LTE FDD	UL Multi-Cluster	ULMultiClusterSwitch option of the CellAlgoSwitch.UlSchExtSwitch parameter	<i>Uplink Scheduling</i>	The number of uplink RBs available for LTE decreases, which reduces the gain provided by UL Multi-Cluster.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Uplink interference randomization	CellUlschAlgo.UlRbAllocationStrategy	<i>ICIC</i>	The number of uplink RBs available for LTE decreases, which reduces the gain provided by uplink interference randomization.
LTE FDD	Uplink frequency selective scheduling	None	<i>Uplink Scheduling</i>	The number of uplink RBs available for LTE decreases, which reduces the gain provided by uplink frequency selective scheduling.
LTE FDD	SRS Configuration Indicator	SRSCfg.SrsCfgI nd	<i>Physical Channel Resource Management</i>	LTE needs to share SRS resources with NR. Therefore, the LTE cell triggers the expansion of the cell-specific SRS subframes in advance, which prolongs the SRS period of LTE UEs.
LTE FDD	Uplink semi-persistent scheduling	SpsSchSwitch option of the CellAlgoSwitch.UlSchSwitch parameter	<i>VoLTE</i>	The number of uplink RBs available for LTE decreases, which reduces the gain provided by uplink semi-persistent scheduling.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	TTI bundling	TtiBundlingSwitch option of the CellAlgoSwitch.UlSchSwitch parameter	<i>VoLTE</i>	The number of uplink RBs available for LTE decreases. Consequently, the maximum number of UEs that can enter the TTI bundling state decreases.
LTE FDD	Uplink short-interval SPS	CellUlSchAlgo.IntvlOfUlSpsWithSkipping	<i>Uplink Scheduling</i>	The number of uplink RBs available for LTE decreases, which reduces the gain provided by uplink short-interval SPS to UEs.
LTE FDD	Uplink MU-MIMO	UlvMimoSwitch option of the CellAlgoSwitch.UlSchSwitch parameter	<i>MIMO</i>	The number of uplink RBs available for LTE decreases, which increases the pairing success rate of uplink MU-MIMO. Consequently, the BLER increases.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	pRRU deep dormancy in intelligent mode	E node B M pru E s .M pru D ormanc y D LEar f cn	<i>Multi-RAT Coordinated Energy Saving</i>	For DBS3900 LampSite and DBS5900 LampSite, when pRRU deep dormancy in intelligent mode is enabled together with LTE FDD and NR Uplink Spectrum Sharing, the demodulation performance of the common PUCCH in NR cells deteriorates.

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
LTE FDD	Physical Resource Slicing	PHY_RES_SLICING_SWITCH option of the CellAlgoExtSwitch.OperatorFunctSwitch parameter	<i>Physical Resource Slicing (FDD)</i>	When LTE FDD and NR Uplink Spectrum Sharing is enabled together with Physical Resource Slicing: - For resource reservation based on the SPID, the number of available PRBs in each PRB group on the LTE side is calculated as follows: (Total spectrum resources – Number of PRBs shared with the non-LTE side) x Proportion of reserved uplink or downlink PRBs for each PRB group. - For resource reservation based on the operator and SPID, the number of available PRBs in each PRB group on the LTE side is calculated as follows: (Total spectrum resources – Number of PRBs shared

RAT	Function Name	Function Switch	Reference (eRAN Feature Documentation)	Description
				with the non-LTE side) x Proportion of reserved uplink or downlink PRBs for each operator x Proportion of reserved uplink or downlink PRBs for each PRB group.

- Function impacts on the NR side

RAT	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
Low-frequency NR TDD High-frequency NR TDD	Uplink interference randomization-based scheduling	NRDUCellAlgoSwitch.UlIntrfRandomSchSwitch	<i>Uplink Scheduling</i>	The number of uplink RBs available for NR is reduced, causing the gain delivered by interference randomization to decrease.

3.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Huawei devices (including the eNodeB and gNodeB) must be used on the radio access network.

Base Station Models

- LTE:
 - 3900 and 5900 series base stations excluding the BTS5929R. 3900 series base stations must be configured with the BBU3900 or BBU3910, and

- 5900 series base stations must be configured with the BBU5900, BBU5910, BookBBU5901a, BookBBU5901, or BBU5900A.
- DBS3900 LampSite and DBS5900 LampSite
- NR:
 - 3900 and 5900 series base stations excluding the BTS5929R. 3900 series base stations must be configured with the BBU3910.
 - DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

- LTE: For baseband processing units, only the UBBPd, UBBPe, UBBPh, and UBBPg support this function. The specific board models and the number of cells supported by each board after function activation are listed in [Table 3-8](#). For main control boards, only the UMPTb, UMPTe, UMPTg, UMPTha, UMPThb, and UMPTga support this function.
- NR: For baseband processing units, only the UBBPfw1, UBBPg2a, UBBPg3, UBBPh, and UBBPg3b support this function. For main control boards, only the UMPTe, UMPTg, UMPTha, UMPThb, and UMPTga support this function.

Table 3-8 Cell number specifications of LTE baseband processing units

Working Mode	Baseband Processing Unit Model	Number of Supported LTE Cells
LTE FDD or LM ^a	UBBPd5	The cell specifications decrease from six 2T2R cells to three cells, or remain three 2T4R or 4T4R cells. All of the three cells can be enabled with LTE FDD and NR Uplink Spectrum Sharing.
LTE FDD or LM	UBBPd6	The cell specifications decrease from six 2T2R, 2T4R, or 4T4R cells to three cells. All of the three cells can be enabled with LTE FDD and NR Uplink Spectrum Sharing.
LTE FDD or LM	UBBPe1	Three 2T2R cells. All of the three cells can be enabled with LTE FDD and NR Uplink Spectrum Sharing.
LTE FDD or LM	UBBPe2	Three 2T2R, 2T4R, or 4T4R cells. All of the three cells can be enabled with LTE FDD and NR Uplink Spectrum Sharing.
LTE FDD or LM	UBBPe3	Six 2T2R cells or three 2T4R or 4T4R cells. Among them, a maximum of three can be enabled with LTE FDD and NR Uplink Spectrum Sharing; other cells are common cells.

Working Mode	Baseband Processing Unit Model	Number of Supported LTE Cells
LTE FDD or LM	UBBP _e 4/UBBP _e 18	Six 2T2R, 2T4R, or 4T4R cells. Among them, a maximum of three can be enabled with LTE FDD and NR Uplink Spectrum Sharing; other cells are common cells.
LTE FDD, LM, or other working modes involving LTE FDD	UBBP _e 5	• LTE FDD or LM: Nine 2T2R, 2T4R, or 4T4R cells. Among them, a maximum of six can be enabled with LTE FDD and NR Uplink Spectrum Sharing; other cells are common cells. • Multimode other than LM: Six 2T2R, 2T4R, or 4T4R cells. Among them, a maximum of three can be enabled with LTE FDD and NR Uplink Spectrum Sharing; other cells are common cells.
LTE FDD, LM, or other working modes involving LTE FDD	UBBP _e 6	• LTE FDD or LM (with BBP.SRT set to DEFAULT): Twelve 2T2R, 2T4R, or 4T4R cells. Among them, a maximum of six can be enabled with LTE FDD and NR Uplink Spectrum Sharing; other cells are common cells. • LM (with BBP.SRT set to NBIOT_ENHANCE): Nine 2T2R, 2T4R, or 4T4R cells. Among them, a maximum of six can be enabled with LTE FDD and NR Uplink Spectrum Sharing; other cells are common cells. • Multimode other than LM: Nine 2T2R, 2T4R, or 4T4R cells. Among them, a maximum of three can be enabled with LTE FDD and NR Uplink Spectrum Sharing; other cells are common cells.
LTE FDD, LM, LN, or other working modes involving LTE FDD	UBBP _g /UBBP _h	Same as that before this function is enabled

Working Mode	Baseband Processing Unit Model	Number of Supported LTE Cells
a: LM refers to LTE FDD and NB-IoT. b: The UBBPe1 is used as an example for illustrating the cell specifications. If this board also serves cells enabled with LTE FDD and NR Flash Dynamic Spectrum Sharing, this board supports a maximum of three cells enabled with LTE FDD and NR Flash Dynamic Spectrum Sharing and enabled with LTE FDD and NR Uplink Spectrum Sharing. The same rule applies to other boards.		

To prevent LTE cell activation failures due to insufficient baseband processing resources, cells enabled with this function must be bound to baseband processing units supporting this function. If cells are not bound to such baseband processing units, cell activation may fail when the baseband processing units supporting this function are fully occupied while other baseband processing units do not support this function.

RF Modules

LTE and NR SUL must share the same RF module. The RRUs listed in [Table 3-9](#) and [Table 3-10](#) support this function.

Table 3-9 RF module requirements (3900 and 5900 series base stations)

Frequency Band	Bandwidth	Module
700 MHz	10 MHz	RRU3262, 5000 series RRUs ^a
800 MHz	10 MHz	RRU3965, 5000 series RRUs ^a
1800 MHz	15 MHz, 20 MHz	RRU3952m, RRU3953, RRU3953w, RRU3959, RRU3959a, RRU3959w, RRU3962, RRU3971, RRU3971a, 5000 series RRUs ^a
2100 MHz	15 MHz, 20 MHz	RRU3952, RRU3952m, RRU3953, RRU3958, RRU3959, RRU3959a, RRU3962, RRU3971, 5000 series RRUs ^a
AWS	15 MHz, 20 MHz	RRU3971, 5000 series RRUs ^a
a: 5000 series RRUs refer to RRUs named RRU5xxx, for example, RRU5901.		

Table 3-10 RF module requirements (DBS3900 LampSite and DBS5900 LampSite)

Frequency Band	Bandwidth	Supported Module
1800 MHz, 2100 MHz	15 MHz, 20 MHz	596X series pRRUs ^a , hRRU5901

Frequency Band	Bandwidth	Supported Module
a: <i>X</i> in pRRU596 <i>X</i> can be a digit or a combination of a digit and a letter, for example, pRRU5961.		

Cells

- 3900 and 5900 series base stations: LTE and NR SUL cells must work on a 10 MHz, 15 MHz, or 20 MHz bandwidth, and have the same operating frequency band, center frequency, and standard bandwidth. The antenna TX/RX mode must be 2T2R, 2T4R, or 4T4R. Sector split cells do not support this function.
- DBS3900 LampSite and DBS5900 LampSite: LTE and NR SUL cells must work on a 10 MHz, 15 MHz, or 20 MHz bandwidth, and have the same operating frequency band, center frequency, and standard bandwidth. The antenna TX/RX mode must be 2T2R. Sector split cells do not support this function.

If LTE FDD and NR Uplink Spectrum Sharing is enabled for an SUL NR DU cell with the **NRDUCellSulRelation.SulType** parameter set to **FIX_SUL**, the **NRDUCell.FrequencyBand** parameter cannot be set to **N89**, **N97**, **N197**, **N198**, or **N199**.

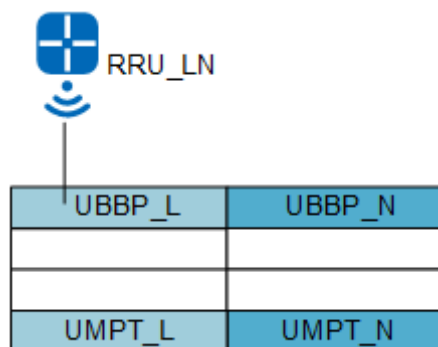
3.3.4 Networking

3900 and 5900 Series Base Stations

In separate-MPT scenarios, this function requires an intra-BBU CPRI MUX or an intra-BBU dual star topology. In co-MPT scenarios, this function requires an intra-BBU load sharing or an intra-BBU CPRI MUX topology. Only 3900 and 5900 series base stations support the intra-BBU dual-star topology in separate-MPT scenarios and intra-BBU load sharing topology in co-MPT scenarios. DBS3900 LampSite and DBS5900 LampSite do not support these topologies.

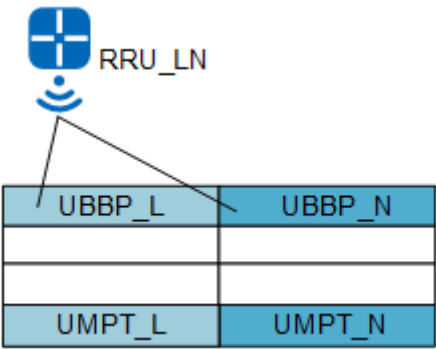
- Figure 3-2** shows an intra-BBU CPRI MUX topology in separate-MPT scenarios. In this topology, LTE and NR share a BBU, and only the LTE baseband processing unit directly connects to the RRU through an optical fiber.

Figure 3-2 Intra-BBU CPRI MUX topology in separate-MPT scenarios



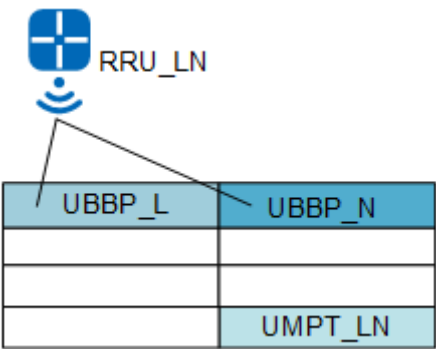
- **Figure 3-3** shows an intra-BBU dual-star topology in separate-MPT scenarios. In this topology, LTE and NR share a BBU, and the LTE and NR baseband processing units connect to the RRU through their own optical fibers.

Figure 3-3 Intra-BBU dual-star topology in separate-MPT scenarios



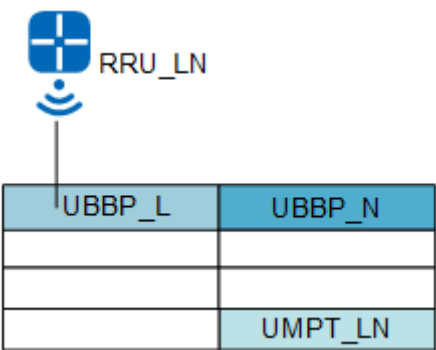
- **Figure 3-4** shows an intra-BBU load sharing topology in co-MPT scenarios. In this topology, LTE and NR share a BBU, and the LTE and NR baseband processing units connect to the RRU through their own optical fibers.

Figure 3-4 Intra-BBU load sharing topology in co-MPT scenarios



- **Figure 3-5** shows an intra-BBU CPRI MUX topology in co-MPT scenarios. In this topology, LTE and NR share a BBU, and only the LTE baseband processing unit directly connects to the RRU through an optical fiber.

Figure 3-5 Intra-BBU CPRI MUX topology in co-MPT scenarios



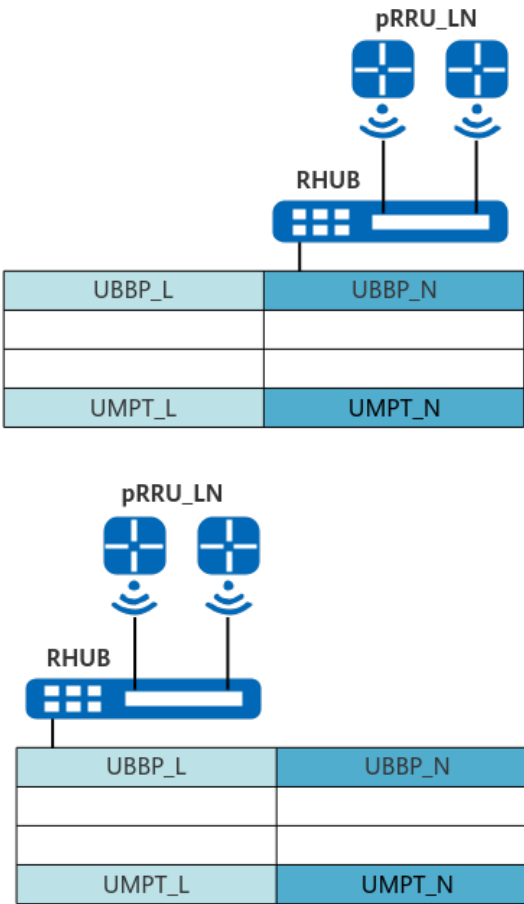
DBS3900 LampSite and DBS5900 LampSite

This function requires an intra-BBU CPRI MUX topology in separate-MPT and co-MPT scenarios. Neither of these topologies poses special requirements on board

installation positions. In either topology, LTE and NR cells can be established on the same baseband processing unit or on different baseband processing units. The following illustrates the networking requirements using LTE and NR cells established on different baseband processing units as an example.

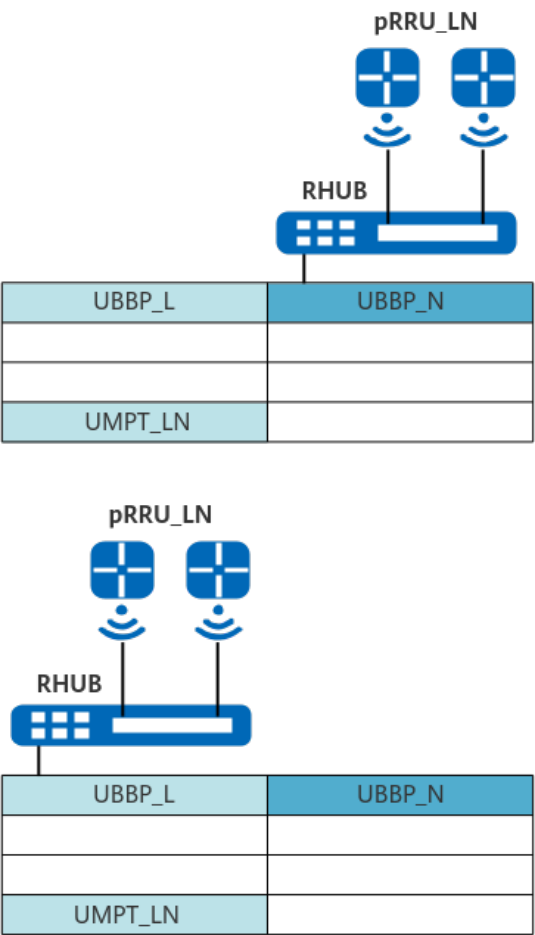
- **Figure 3-6** shows an intra-BBU CPRI MUX topology in separate-MPT LTE/NR scenarios. In this topology, LTE and NR share a BBU, and only the LTE baseband processing unit or NR baseband processing unit directly connects to the RHUB through an optical fiber.

Figure 3-6 Intra-BBU CPRI MUX topology in separate-MPT LTE/NR scenarios



- **Figure 3-7** shows an intra-BBU CPRI MUX topology in co-MPT LTE/NR scenarios. In this topology, LTE and NR share a BBU, and only the LTE baseband processing unit or NR baseband processing unit directly connects to the RHUB through an optical fiber.

Figure 3-7 Intra-BBU CPRI MUX topology in co-MPT LTE/NR scenarios



For the DBS3900 LampSite and DBS5900 LampSite, LTE and NR cells support only one sector equipment group. If an RHUB5965 is used, each sector equipment group supports a maximum of 24 pRRUs. If another RHUB model is used, each sector equipment group supports a maximum of 16 pRRUs.

3.3.5 Others

- Both MRFD-131223 LTE FDD and NR Uplink Spectrum Sharing (LTE FDD) and MRFD-131263 LTE FDD and NR Uplink Spectrum Sharing (NR) must be enabled.
- If the LTE and NR co-carrier co-CPRI data function and CPRI compression function are both enabled, the CPRI compression ratios must be the same between the LTE and NR SUL cells. The CPRI compression type of LTE cells is specified by the **Cell.CPRICompression** parameter, and the CPRI compression type of NR SUL cells is specified by the **NRDUCellTrp.CpriCompression** parameter. [Table 3-11](#) describes the configuration requirements.

Table 3-11 Configuration requirements on the CPRI compression types of the LTE and NR SUL cells

Bandwidth	CPRI Compression Ratio	CPRI Compression Type for LTE Cells	CPRI Compression Type for NR SUL Cells
10 MHz	No CPRI compression	NO_COMPRESSION	NO_COMPRESSION
	3:2 CPRI compression	NORMAL_COMPRESSION	1DOT5_COMPRESSION
15 MHz/20 MHz	No CPRI compression	NO_COMPRESSION	NO_COMPRESSION
	2:1 CPRI compression	NORMAL_COMPRESSION	2_COMPRESSION

- When this function is used together with uplink FDD+TDD CA or downlink FDD+TDD CA:
 - If LTE TDD and NR TDD (SUL) are deployed in the same frequency band, and if LTE FDD and NR Uplink Spectrum Sharing is enabled together with UL and DL Decoupling or Super Uplink in the LTE FDD frequency band, LTE FDD and NR Uplink Spectrum Sharing is mutually exclusive with FDD+TDD CA.
 - If LTE TDD and NR TDD (SUL) are deployed in different frequency bands, and if LTE FDD and NR Uplink Spectrum Sharing is enabled together with UL and DL Decoupling or Super Uplink in the LTE FDD frequency band, LTE FDD and NR Uplink Spectrum Sharing can be used together with FDD+TDD CA as long as the following conditions are met:
 - The effective value of LTE TDD cell frame offset (specified by the **CellFrameOffset.FrameOffset** or **ENodeBFrameOffset.TddFrameOffset** parameter. If these parameters are both configured, the value of the **CellFrameOffset.FrameOffset** parameter prevails.) is identical with the effective value of NR TDD cell frame offset (specified by the **gNodeBParam.FrameOffset** parameter).
 - The effective value of the frame offset of the LTE TDD cell is greater than that of the LTE FDD cell (specified by the **CellFrameOffset.FrameOffset** or **ENodeBFrameOffset.FddFrameOffset** parameter). The difference between these two values is identical with the effective value of the TA offset of the NR TDD cell (specified by the **NRDUCell.TaOffset** parameter). [Table 3-12](#) describes the configuration requirements.

Table 3-12 Configuration requirements on the difference between the frame offset effective values of LTE TDD and LTE FDD cells and on the TA offset effective value of NR TDD cells

Difference Between the Frame Offset Effective Values of LTE TDD and LTE FDD Cells ^a	Effective Value of NR TDD Cell TA Offset	Other Restriction
624Ts	39936Tc	The distance between the antennas of the LTE FDD and LTE TDD cells must be less than or equal to 2.9 km. Otherwise, the proportion of UEs for which uplink FDD+TDD CA takes effect decreases.
400Ts	25600Tc	The distance between the antennas of the LTE FDD and LTE TDD cells must be less than or equal to 5.1 km. Otherwise, the proportion of UEs for which uplink FDD+TDD CA takes effect decreases.
a: Difference between the frame offset effective values of LTE TDD and LTE FDD cells = Frame offset effective value of LTE TDD cells – Frame offset effective value of LTE FDD cells		

The configurations of LTE FDD, LTE TDD, and NR TDD cell frame offsets and the NR TDD cell TA offset must be confirmed with Huawei engineers.

3.4 Operation and Maintenance

3.4.1 When to Use

This function can be used in LTE cells with a light uplink load (uplink PRB usage < 30%), since it decreases the amount of uplink spectrum available for LTE. The uplink PRB usage of an LTE cell can be obtained by dividing **L.ChMeas.PRB.UL.Used.Avg** by **L.ChMeas.PRB.UL.Avail**. You are advised to use the average uplink PRB usage in busy hours.

3.4.2 Data Configuration

NOTICE

If the activation of this function is to be verified through signaling message tracing, you need to start signaling tracing on the MAE-Access before activating this function in LTE and NR SUL cells. Otherwise, activation-related messages cannot be observed after this function is activated.

3.4.2.1 Data Preparation

Table 3-13 describes the parameters used for function activation on the LTE side. **Table 3-14** describes the parameters used for function activation on the NR side. This section does not describe parameters related to cell establishment.

Since the configuration of some of these parameters can result in automatic cell restarts, it is recommended that you deactivate the cells prior to parameter configuration and reactivate them after parameter configuration is complete. This will prevent these cells from repeatedly restarting during parameter configuration.

Table 3-13 LTE parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Spectrum Cloud Switch	SpectrumCloud.SpectrumCloudSwitch	Set this parameter to LTE_NR_UPLINK_SPECTRUM_SHR .
LTE and NR Spectrum Sharing Cell Group ID	SpectrumCloud.LteNrSpectrumShrCellGrpId	Set this parameter based on the network plan.
Spectrum Cloud Enhancement Switch	SpectrumCloud.SpectrumCloudEnhSwitch	Select the LNR_SPECTRUM_SHR_SUPERUL_SW option of this parameter.
LTE and NR Spectrum Sharing Cell Group ID	LteNrSpctShrCellGrp.LteNrSpectrumShrCellGrpId	Set this parameter based on the network plan. The value of this parameter must be the same as that of the gNBDULteNrSpctShrCg.LteSpctShrCellGrpId parameter.

Parameter Name	Parameter ID	Setting Notes
LTE Pri Res Ratio in LTE and NR Spct Shr	LteNrSpctShrCellGrp.LteNrSpctShrLtePriResRatio	Set this parameter based on the network plan. This parameter value varies with the spectrum allocation policy. The following lists the configuration suggestions for ensuring the network performance when different spectrum allocation policies are adopted: • If spectrum resources need to be preferentially allocated to LTE, it is recommended that this parameter be set to a value in the range of 51 to 80. • If spectrum resources need to be preferentially allocated to NR, it is recommended that this parameter be set to a value in the range of 20 to 49. • If spectrum resources need to be evenly allocated to LTE and NR, you are advised to set this parameter to 50.
RB Reserve Mode ^a	CellRbReserve.RbRsvMode	Set this parameter to LNR_SPECTRUM_SHR_RB_RESERVED when channel decoupling is enabled on the NR side.
RB Reserve Type ^a	CellRbReserve.RbRsvType	Set this parameter to UPLINK_MODE when channel decoupling is enabled on the NR side.

Parameter Name	Parameter ID	Setting Notes
RB Reserve Start Index ^a	CellRbReserve.RbRsvStartIndex	Set this parameter to 0 when channel decoupling is enabled on the NR side.
RB Reserve End Index ^a	CellRbReserve.RbRsvEndIndex	Set this parameter as follows when channel decoupling is enabled on the NR side: - Set this parameter to 49 in 10 MHz bandwidth scenarios. - Set this parameter to 74 in 15 MHz bandwidth scenarios. - Set this parameter to 99 in 20 MHz bandwidth scenarios.
LTE and NR Co Carrier Co CPRI Data Switch	LteNrSpctShrCellGrp.LteNrCoCarrCoCpriDataSw	Set this parameter based on the network plan. This parameter can be set to ON only when LTE and NR use the CPRI MUX topology.
Frame Offset ^b	CellFrameOffset.FrameOffset	Set this parameter to a value confirmed by Huawei engineers.
FDD Frame Offset ^b	ENodeBFrameOffset.FddFrameOffset	Set this parameter to a value confirmed by Huawei engineers.
Clock Working Mode ^c	TASM.MODE	Set this parameter to MANUAL .
Selected Clock Source ^c	TASM.CLKSRC	Set this parameter to PEERCLK .
Peer Clock No. ^c	PEERCLK.PN	Set this parameter based on the network plan.
Peer Clock Node Type ^c	PEERCLK.PNT	Set this parameter to BBU .
Peer Standard ^c	PEERCLK.PS	Set this parameter to NR .
Clock Synchronization Mode ^c	TASM.CLKSYNCMODE	Set this parameter to TIME .

Parameter Name	Parameter ID	Setting Notes
a: If channel decoupling is not enabled on the NR side, the four parameters do not need to be configured. When modifying the LTE cell uplink bandwidth, remove the frequency-domain configuration of the LTE PUCCH at the lower and upper ends if the LTE PUCCH has been configured. That is, remove the settings of the CellRbReserve.RbRsvMode, CellRbReserve.RbRsvType, CellRbReserve.RbRsvStartIndex, and CellRbReserve.RbRsvEndIndex parameters. b: If both the frame offset (specified by the CellFrameOffset.FrameOffset parameter) and the FDD frame offset (specified by the ENodeBFrameOffset.FddFrameOffset parameter) are configured, the frame offset (specified by the CellFrameOffset.FrameOffset parameter) takes effect for LTE cells. c: In separate-MPT LTE/NR scenarios, if time synchronization needs to be reconfigured on the LTE side according to the network plan, these six parameters need to be configured.		

Table 3-14 NR parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Spectrum Cloudification Switch	NRDUCellAlgoSwitch.SpectrumCloudSwitch	Select the LTE_NR_UL_SPECTRUM_SHARING_SW option of this parameter.
NR Spectrum Sharing Cell Group ID	NRDUCellSpctCloud.NrSpctShrCellGrpId	Set this parameter based on the network plan.
NR Spectrum Sharing Cell Group ID	gNBDULteNrSpctShrCg.NrSpctShrCellGrpId	Set this parameter based on the network plan.
LTE Spectrum Sharing Cell Group ID	gNBDULteNrSpctShrCg.LteSpctShrCellGrpId	Set this parameter based on the network plan. The value of this parameter must be the same as that of the LteNrSpctShrCellGrp.LteNrSpectrumShrCellGrpId parameter.
Frame Offset ^a	gNodeBParam.FrameOffset	Set this parameter to a value confirmed by Huawei engineers.
Frame Offset ^a	gNBFreqBandConfig.FrameOffset	Set this parameter to a value confirmed by Huawei engineers.

Parameter Name	Parameter ID	Setting Notes
a: If both the gNodeBParam.FrameOffset and gNBFreqBandConfig.FrameOffset parameters are configured, the frame offset specified by the gNBFreqBandConfig.FrameOffset parameter takes effect for NR SUL cells.		

3.4.2.2 Using MML Commands

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Before using MML commands, refer to [3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

Activation command examples on the LTE side

```
//Deactivating a cell
DEA CELL: LocalCellId=0;
//Binding a cell that requires this function to a baseband processing unit supporting this function
MOD EUCELLSECTOREQM: LocalCellId=0, SectorEqmId=1, BaseBandEqmId=12;
//Optional, required only when the primary baseband equipment is configured for a cell Binding a cell
that requires this function to the primary baseband equipment supporting this function
MOD EUCELLPRIBBEQM: LocalCellId=0, PriBaseBandEqmId=12;
//Setting the percentage of spectrum resources to be preferentially allocated to LTE services, and turning on
the LTE and NR co-carrier co-CPRI data switch (optional, set based on the network plan)
ADD LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0, LteNrSpectShrLtePriResRatio=50,
LteNrCoCarrCoCpriDataSw=ON;
//Turning on the LTE and NR uplink spectrum sharing switch and adding an LTE cell to an LTE spectrum
sharing cell group
ADD SPECTRUMCLOUD: LocalCellId=0, SpectrumCloudSwitch=LTE_NR_UPLINK_SPECTRUM_SHR,
LteNrSpectrumShrCellGrpId=0;
//Turning on the super uplink spectrum sharing switch
MOD SPECTRUMCLOUD: LocalCellId=0, SpectrumCloudEnhSwitch=LNR_SPECTRUM_SHR_SUPERUL_SW-1;
//Optional, required only when channel decoupling is enabled on the NR side Configuring the start
position of the LTE PUCCH at the lower end and the end position of the LTE PUCCH at the upper end. The
following uses a 10 MHz cell as an example.
ADD CELLRBRESERVE: LocalCellId=0, Index=1, RbRsvMode=LNR_SPECTRUM_SHR_RB_RESERVED,
RbRsvType=UPLINK_MODE, RbRsvStartIndex=0, RbRsvEndIndex=49;
//Setting the frame offset
ADD CELLFRAMEOFFSET: LocalCellId=0, FrameOffsetMode=CustomFrameOffset, FrameOffset=306800;
//Optional, required only in separate-MPT LTE/NR base stations Configuring time synchronization as per
the network plan
ADD PEERCLK: PN=0, PNT=BBU, PS=NR;
SET CLKMODE: MODE=MANUAL, CLKSRC=PEERCLK;
SET CLKSYNCMODE: CLKSYNCMODE=TIME;
//Activating a cell
ACT CELL: LocalCellId=0;
```

Activation command examples on the NR side

```
//Deactivating the NR TDD cell that is associated with the NR SUL cell
DEA NRCELL: NrCellId=0;
//Adding the association between an LTE spectrum sharing cell group and an NR spectrum sharing cell
```

```
group
ADD GNBDULTENRSPCTSHRCG: NrSpctShrCellGrpId=0, LteSpctShrCellGrpId=0;
//Adding an NR SUL cell to the NR spectrum sharing cell group
ADD NRDUCELLSPCTCLOUD: NrDuCellId=1, NrSpctShrCellGrpId=0;
//Setting the frame offset
MOD GNODEBPARAM: FrameOffset=0;
//Enabling LTE FDD and NR Uplink Spectrum Sharing
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
SpectrumCloudSwitch=LTE_NR_UL_SPECTRUM_SHARING_SW-1;
//Activating the NR TDD cell that is associated with the NR SUL cell
ACT NRCELL: NrCellId=0;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

Deactivation command examples on the LTE side

```
//Deactivating a cell
DEA CELL: LocalCellId=0;
//Removing the RB reservation for LTE and NR spectrum sharing, if configured
RMV CELLRBRESERVE: LocalCellId=0, Index=1;
//Turning off the LTE and NR uplink spectrum sharing switch and super uplink spectrum sharing switch
MOD SPECTRUMCLOUD: LocalCellId=0, SpectrumCloudSwitch=OFF,
SpectrumCloudEnhSwitch=LNR_SPECTRUM_SHR_SUPERUL_SW-0;
//Removing an LTE cell from the LTE spectrum sharing cell group
RMV SPECTRUMCLOUD: LocalCellId=0;
RMV LTENRSPCTSHRCELLGRP: LteNrSpectrumShrCellGrpId=0;
//Activating a cell
ACT CELL: LocalCellId=0;
```

Deactivation command examples on the NR side

```
//Deactivating the NR TDD cell that is associated with the NR SUL cell
DEA NRCELL: NrCellId=0;
//Disabling LTE FDD and NR Uplink Spectrum Sharing
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
SpectrumCloudSwitch=LTE_NR_UL_SPECTRUM_SHARING_SW-0;
//Removing an NR SUL cell from the NR spectrum sharing cell group
RMV NRDUCELLSPCTCLOUD: NrDuCellId=1;
RMV GNBDULTENRSPCTSHRCG: NrSpctShrCellGrpId=0;
//Activating the NR TDD cell that is associated with the NR SUL cell
ACT NRCELL: NrCellId=0;
```

3.4.2.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

3.4.3 Activation Verification

Check whether this function has taken effect by using MML commands, observing counters, or tracing signaling messages.

Using MML Commands

The LTE and NR uplink spectrum sharing function takes effect when both of the following conditions are met:

- The value of the **Spectrum Sharing Status** parameter in the output of the eNodeB MML command **DSP LTENRSPCTSHRCELLGRP** is **SUPER_UL_DYNAMIC_SHARING**.
- The value of the **Spectrum Sharing Status** parameter in the output of the gNodeB MML command **DSP GNBDULTENRSPCTSHRCG** is **Dynamic Sharing**.

Observing Counters

The LTE and NR uplink spectrum sharing function takes effect when the following conditions for eNodeB and gNodeB are all met:

- Observe the following counter on the eNodeB side:
L.ChMeas.PRB.UL.Actual.Avail (counter ID: 1526759050): This counter measures the number of actually available uplink PRBs and its value is less than the number of RBs corresponding to the cell bandwidth specified by the **Cell.UlBandWidth** parameter.
- Observe the following counter on the gNodeB side:
N.PR.B.UL.Actual.Avail.Avg (counter ID: 1911827161): This counter measures the average number of actually available uplink PRBs and its value is less than the number of RBs corresponding to the cell bandwidth specified by the **NRDUCell.UlBandwidth** parameter.

Tracing Signaling Messages

NOTICE

Before activating this function in LTE and NR SUL cells, start signaling tracing on the MAE-Access. Otherwise, activation-related messages cannot be observed after this function is activated.

The LTE and NR uplink spectrum sharing function takes effect when the following conditions for eNodeB and gNodeB are all met:

Perform the following steps to start eNodeB message tracing on the MAE-Access:

- Step 1** Log in to the MAE-Access. Choose **Monitor > Signaling Trace > Signaling Trace Management**. In the navigation tree of the displayed window, expand **LTE > Application Layer > Inter-RAT Huawei-Proprietary Interface Trace**.
- Step 2** In the displayed dialog box, select an NE and set related parameters. Then, click **Finish** to start an inter-RAT Huawei-proprietary interface tracing task.
- Step 3** After this function is activated in LTE and NR SUL cells, messages are traced over the inter-RAT Huawei-proprietary interface. View the value of the **lnss-group-status-change-info > nr-ul-cell-information > spectrum-sharing-status** IE in the **LNR_INTER_RAT_SPECTRUM_CHANGE_INDICATION** message. The value is expected to be **ss-dynamic-sharing**.

----End

Perform the following steps to start gNodeB message tracing on the MAE-Access:

- Step 1** Log in to the MAE-Access. Choose **Monitor > Signaling Trace > Signaling Trace Management**. In the navigation tree of the displayed window, expand **NR > Application Layer > Inter-RAT Huawei-Proprietary Interface Trace**.
- Step 2** In the displayed dialog box, select an NE and set related parameters. Then, click **Finish** to start an inter-RAT Huawei-proprietary interface tracing task.
- Step 3** After this function is activated in LTE and NR SUL cells, messages are traced over the inter-RAT Huawei-proprietary interface. View the value of the **lnss-group-status-change-info > nr-ul-cell-information > spectrum-sharing-status** IE in the **LNR_INTER_RAT_SPECTRUM_CHANGE_INDICATION** message. The value is **ss-dynamic-sharing**.
- End

3.4.4 Network Monitoring

The LTE uplink throughput decrease can be monitored using the counters listed in [3.2.2 Impacts](#). The NR SUL cell uplink throughput increase is described in the "Network Monitoring" section in *Super Uplink* or in *UL and DL Decoupling*.

4 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

5 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

6 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

7 Reference Documents

- Feature parameter description documents in *eRAN Feature Documentation*:
 - *Carrier Aggregation*
 - *Uplink Scheduling*
 - *ICIC*
 - *Extended CP*
 - *Physical Channel Resource Management*
 - *MIMO*
 - *Random Access Control*
 - *SFN*
 - *High Speed Mobility*
 - *RuralLink Solution*
 - *Compact Bandwidth (FDD)*
 - *Flexible Bandwidth based on Overlap Carriers (FDD)*
 - *GSM and LTE Buffer Zone Optimization*
 - *UL Refarming Zero Bufferzone*
 - *CDMA and LTE Zero Bufferzone*
 - *UMTS and LTE Zero Bufferzone*
 - *Relay*
 - *eMTC*
 - *VoLTE*
 - *NB-IoT Basics (FDD)*
 - *Superior Uplink Coverage (FDD)*
 - *Uplink Coordinated Scheduling*
 - *CSPC*
 - *Smart 8T8R (FDD)*
 - *Seamless Intra-Band Carrier Joining (FDD)*
 - *PTT*
 - *Short TTI (FDD)*

- *Energy Conservation and Emission Reduction*
 - *Physical Resource Slicing (FDD)*
 - *Interference Detection and Suppression*
 - *LTE Flexible Spectrum (FDD)*
- Feature parameter description documents in *5G RAN Feature Documentation*:
 - *Super Uplink*
 - *Channel Management*
 - *QoS Management*
 - *Admission Control*
 - *Indoor Coverage*
 - *Uplink Scheduling*
 - *LumiLink Solution*
- *LTE Spectrum Coordination*
- *GSM and LTE Spectrum Concurrency*
- *UMTS and LTE Spectrum Sharing*
- *UMTS and LTE Spectrum Sharing Based on DC-HSDPA*
- *UL and DL Decoupling*
- *LTE FDD and NR Dynamic Spectrum Sharing*